

DESIGN DOCUMENT FOR HW-5

GROUP-31

Q : Brief explanation of implementation

Ans : `recv_message` handles messages received by the server from a client. It interprets the message and forwards it to the correct recipients using `send_message`. It supports two types of messages from clients:

1. LIST – request for a list of currently connected clients.
2. DATA – message to one or more other clients.

LIST

When a client sends the LIST command, the server responds with a list of all currently connected clients, excluding the sender. It scans the `valid_ids` array to identify active clients and builds a formatted string showing their IDs. This allows clients to know who they can send messages to. The response is sent back to the requesting client using the `send_message` function

DATA

When a client sends a DATA message, it specifies the number of recipients, their client IDs, and the message content (e.g., `DATA 3 1 2 3: Hello`). The server parses the message, validates each recipient ID, and forwards the message to all valid recipients via `send_message`. The server also logs the original message for monitoring. This enables inter-client communication while ensuring that messages are only sent to connected clients.

Q-1 What is the output on all clients' and the server's terminals when the fifth client enters

LIST

Q-2 What is the output on all clients' and the server's terminals when the fifth client enters

DATA 3 3 1 2: hello world

Q-3 What is the output on all clients' and the server's terminals when the fifth client enters

DATA 1 0: hello world

SERVER

```
saksham@SAKSHAMS-PC:~$ cd ~/CN/hw_cs
saksham@SAKSHAMS-PC:~/CN/hw_cs$ ./server
No activity for 30 secs.
client 0 connected
client 1 connected
client 2 connected
client 3 connected
client 4 connected
client 5 connected
client 5: DATA 3 3 1 2: hello world
client 5: DATA 1 0: hello world
No activity for 30 secs.
No activity for 30 secs.
No activity for 30 secs.
```

CLIENT-5

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CN/hw_cs
saksham@SAKSHAMS-PC:~/CN/hw_cs$ ./client
LIST
server: Connected clients: 0 1 2 3 4
DATA 3 3 1 2: hello world
DATA 1 0: hello world
No activity for 30 secs.
No activity for 30 secs.
```

CLIENT-0

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CN/hw_cs
saksham@SAKSHAMS-PC:~/CN/hw_cs$ ./client
No activity for 30 secs.
client-5: hello world
No activity for 30 secs.
```

CLIENT-1

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CN/hw_cs
saksham@SAKSHAMS-PC:~/CN/hw_cs$ ./client
No activity for 30 secs.
client-5: hello world
```

CLIENT-2

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CN/hw_cs
saksham@SAKSHAMS-PC:~/CN/hw_cs$ ./client
No activity for 30 secs.
client-5: hello world
```

CLIENT-3

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CN/hw_cs
saksham@SAKSHAMS-PC:~/CN/hw_cs$ ./client
No activity for 30 secs.
client-5: hello world
```

CLIENT-4

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CN/hw_cs
saksham@SAKSHAMS-PC:~/CN/hw_cs$ ./client
No activity for 30 secs.
No activity for 30 secs.
```

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